Interactive Web Pages

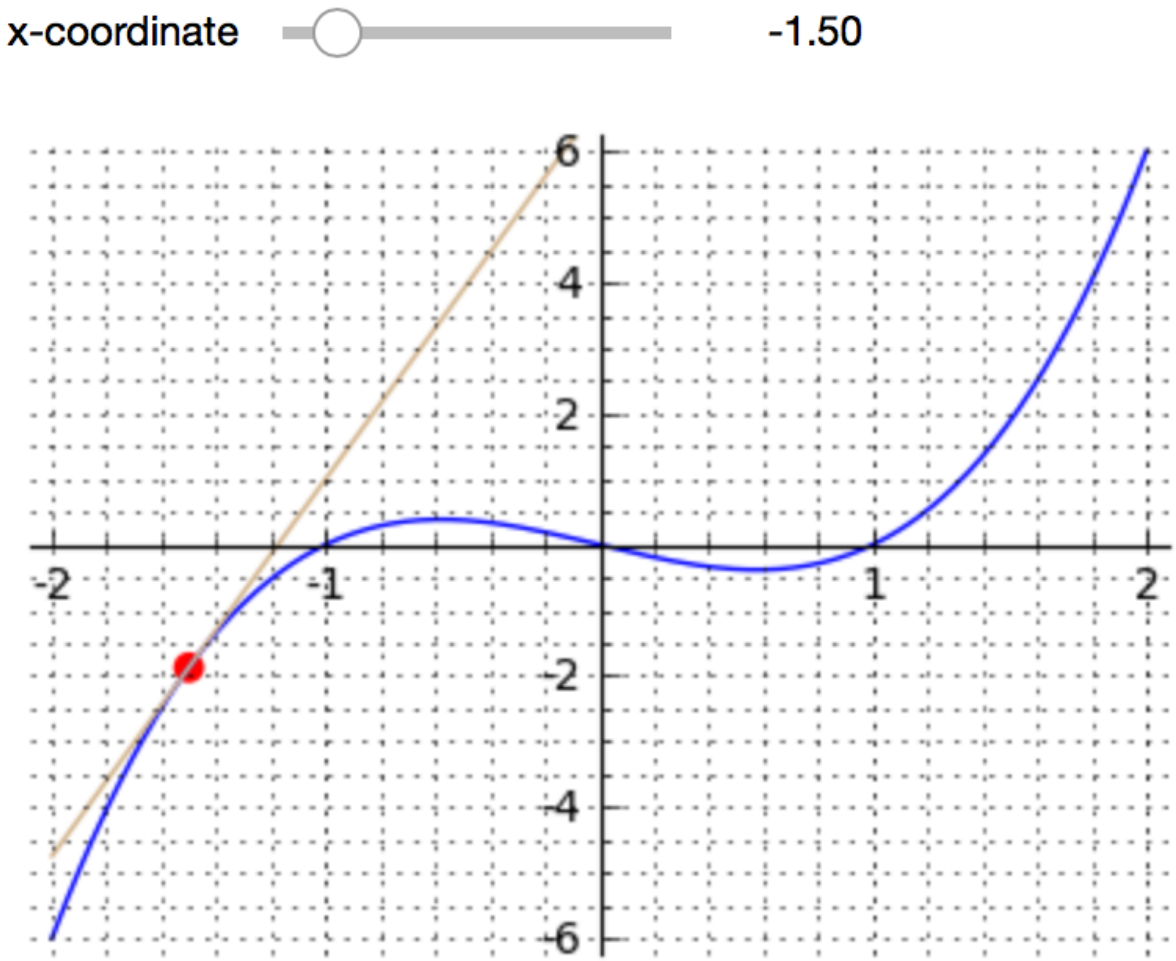
1. Plots with a Parameter
   - drawing tangent lines to a curve
   - elements of interacts
2. An Application
   - a four bar mechanism
   - solving a system of polynomial equations

MCS 320 Lecture 32
Introduction to Symbolic Computation
Jan Verschelde, 6 November 2026

Interactive Web Pages

1. Plots with a Parameter
   - drawing tangent lines to a curve
   - elements of interacts
2. An Application
   - a four bar mechanism
   - solving a system of polynomial equations

Tangent Lines to a Curve



The *x*-coordinate of the point at the curve is the parameter we move.

Interactive Web Pages

1. Plots with a Parameter
   - drawing tangent lines to a curve
   - elements of interacts
2. An Application
   - a four bar mechanism
   - solving a system of polynomial equations

Elements of Interacts

Three stages in building an interactive web page.

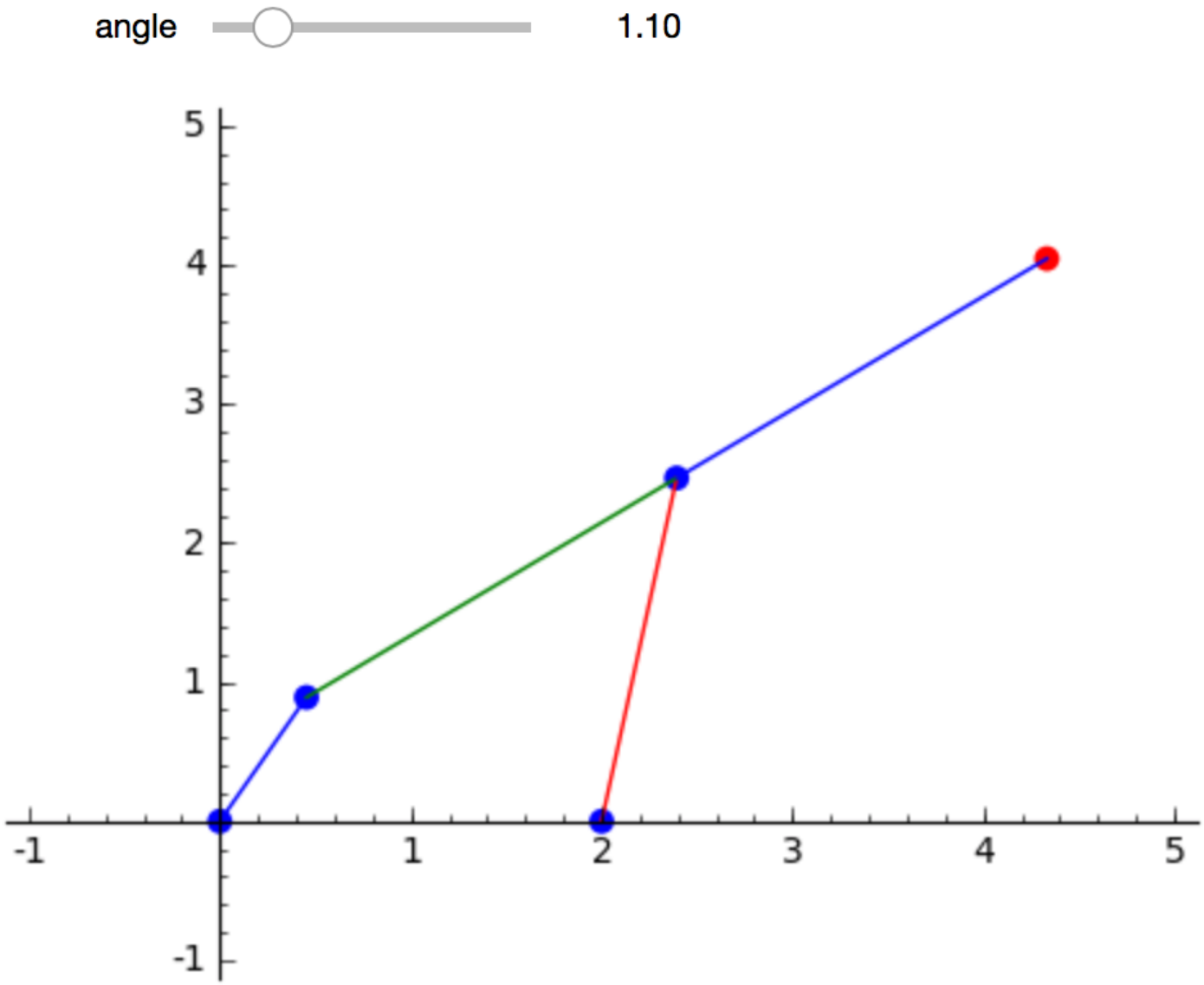
1. Define the moving parameter in the plot.
2. Similar to a Graphical User Interface,
   we have sliders, selectors, drop down menus.
3. Deploy the interact on an HTML page.

Interactive Web Pages

1. Plots with a Parameter
   - drawing tangent lines to a curve
   - elements of interacts

2. An Application
   - a four bar mechanism
   - solving a system of polynomial equations

A Four Bar Mechanism



Interactive Web Pages

1. Plots with a Parameter
   - drawing tangent lines to a curve
   - elements of interacts

2. An Application
   - a four bar mechanism
   - solving a system of polynomial equations

The Connector Point

The coordinates (x, y) of the connector point are at the intersection of two circles.

1. The point (x, y) is at distance $5/2$ from the point $(2, 0)$:

$$(x-2)^2 + y^2 - \left(\frac{5}{2}\right)^2 = 0.$$

2. The point (x, y) is at distance $5/2$ from the point (c, s):

$$(x-c)^2 + (y-s)^2 - \left(\frac{5}{2}\right)^2 = 0,$$

   where $c = \cos(\theta)$ and $s = \sin(\theta)$.

With resultants we compute expressions for (x, y) in function of the parameters c and s.